

1.

$$\delta = \frac{PL}{AE}$$

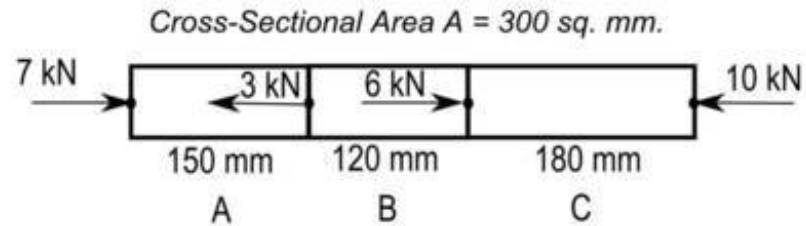
Which of the listed choices (A - H) are used to develop the expression shown for delta, the deflection of a uniform, axially loaded member ?
(Select all that are required)

- A. Hooke's Law
- B. Poisson's Ratio
- C. Definition of Normal Strain
- D. Coefficient of Thermal Expansion
- E. Definition of Normal Stress
- F. Definition of Shear Stress
- G. Principle of Superposition
- H. Strain Energy

2.

A uniform aluminum rod ($E = 70 \text{ GPa}$) has a cross-sectional area $A = 300 \text{ square millimeters}$ and a total length of 450 mm . It is subject to axial loads as shown.

By how much does the center section (B) deflect? Submit your answer in millimeters with a (+) for lengthening or a (-) for shortening.



3.

A rectangular bar is axially loaded in z direction with force P (N). The lateral dimensions a (mm) and b (mm) are along x and y directions. The material is homogeneous and isotropic. Poisson's ratio ν is

A. $-\epsilon_x/\epsilon_y$

B. $-\epsilon_y/\epsilon_x$

C. $-\epsilon_x/\epsilon_z$

D. $-\epsilon_y/\epsilon_z$

E. Both (a) and (b)

F. Both (c) and (d)

G. None

4.

Poisson's ratio of Rubber is

A. 0

B. 0.2

C. 0.32

D. 0.5

5.

If a cylindrical bar of length L and diameter D and an elastic modulus of E is axially loaded with a force P . The Poisson's ratio ν is given by

- A. $\delta D / D$
- B. $-(\delta D / D) / E$
- C. $-(\delta D / D) / (P / AE)$
- D. $+(\delta D / D) / (\sigma / E)$

6.

For an axially loaded bar, the maximum shear stress is along the following planes

- A. planes perpendicular to the axial force
- B. Planes parallel to the axial force
- C. Places 45° to the axial force

7.

We have a $L=1\text{m}$ long cylindrical steel bar with diameter $D=10\text{mm}$. The elastic modulus of steel is $E=200\text{ GPa}$ and the Poisson's ratio is $\nu = 0.32$. If we apply a compressive axial force of $P=5\text{ kN}$, what is the change in diameter δD in mm?

- A. 0.00102 mm
- B. 4 mm
- C. 0.205 mm
- D. 0.04 mm

8.

A cylindrical bar of a metal alloy $D = 20$ mm, length $L = 200$ mm is deformed elastically in tension. The axial force is $F = 50,000$ N. Young's modulus $E = 70$ GPa, Poisson's ratio $\nu = 0.34$.

What is the change in length δL in millimetres?

9.

A cylindrical bar of a metal alloy $D = 20$ mm, length $L = 200$ mm is deformed elastically in tension. The axial force is $F = 50,000$ N. Young's modulus $E = 70$ GPa, Poisson's ratio $\nu = 0.34$.

What is the change in diameter δD in millimetres?

Note: contraction is negative , extension is positive.

10.

A brittle material is under uniaxial tensile loading. A small elliptical surface crack is observed on one side of the specimen. Two important formulas are:

Stress concentration factor: $K = \sigma_{\max} / \sigma_{\text{nominal}}$

Griffith's critical stress for crack propagation:

$$\sigma_c = \{ 2 E \gamma / \pi a \}^{0.5}$$

Where the σ_c : critical stress for crack growth

E: Young's modulus

γ : specific surface energy,

a : half crack length

Which of the following statement is most accurate?

- A. The stress concentration factor predicts the exact stress at which a crack will propagate in a brittle material
- B. Griffith's formula applies to ductile materials and considers plastic deformation near the crack tip.
- C. The presence of a sharp crack increases the local stress, and if this exceeds the critical stress from the Griffith's formula, crack growth occurs.
- D. The nominal stress is always equal to the maximum stress near a crack tip.